

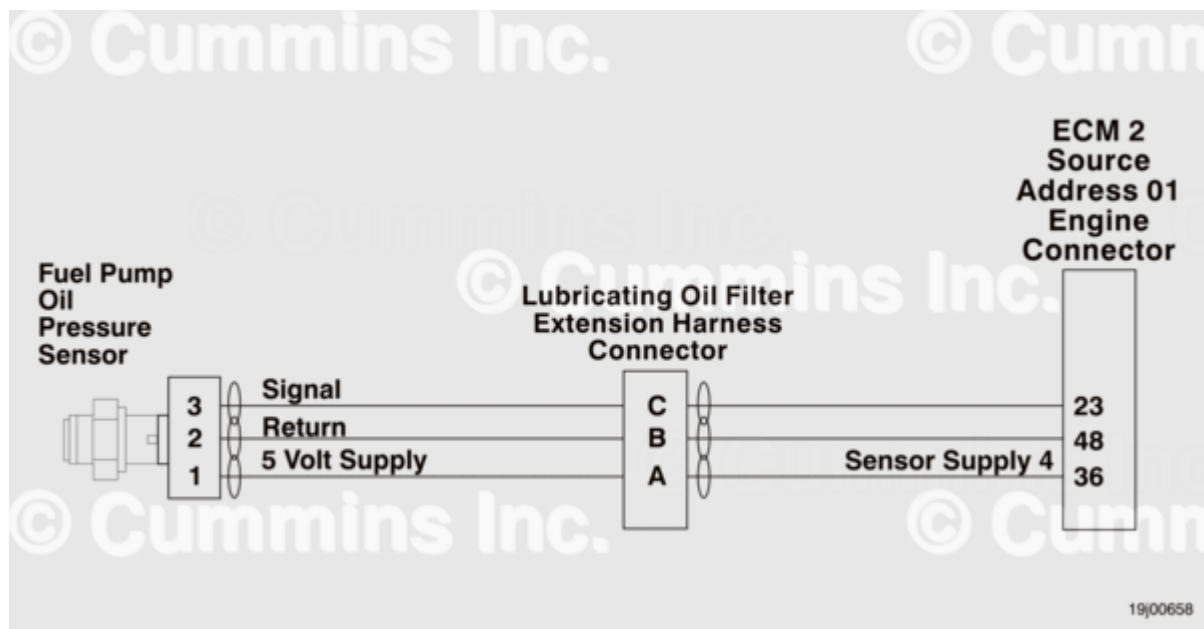
Fault Code: 5121

Fuel Pump Oil Pressure Sensor Circuit - Voltage Below Normal or Shorted to Low Source

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Overview

Codes	Reason	Effect
Fault Code: 5121 PID(P): SID(S) SPN: 520754 FMI: 4/4 Lamp: Amber SRT:	Low signal voltage or open circuit detected at the fuel pump oil pressure circuit.	None on performance.



Engine Fuel Pump Oil Pressure Sensor Circuit

Circuit Description

The engine control module (ECM) provides a 5-volt supply to the engine fuel pump oil pressure sensor on the sensor

supply circuit. The ECM also provides a ground on the sensor return circuit. The engine fuel pump oil pressure sensor

provides a signal to the ECM on the engine fuel pump oil pressure sensor signal circuit. This sensor signal voltage

changes based on the pressure in the oil flow path. The ECM will detect a low signal voltage at operating conditions

when the oil pressure may be slightly lower. The ECM will detect a high signal voltage during high engine speeds or

operating conditions when the oil pressure is high.

Component Location

The engine fuel pump oil pressure sensor is located in the engine fuel pump oil filter head mounted to the top of the

fuel pump adapter drive. Certain engines may have the filter head mounted remotely.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the keyswitch is in the ON position.

Conditions For Setting The Fault Codes

Action Taken When The Fault Code Is Active

- The ECM illuminates the amber CHECK ENGINE lamp immediately when the diagnostic runs and fails.
- A default value for the fuel pump oil pressure reading is used.

Conditions For Clearing The Fault Code

- To validate the repair, perform a key cycle, start the engine and let it idle for 1 minute.
- The fault code status displayed by the recommended Cummins® electronic service tool or equivalent will change to INACTIVE immediately after the diagnostic runs and passes.
- The ECM will turn off the amber CHECK ENGINE lamp immediately after the diagnostic runs and passes.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active and inactive faults.

Shoptalk

Possible causes of this fault code include:

- Signal circuit open or shorted to ground in the engine wiring harness or sensor.

- Supply wire open or shorted to ground.

Refer to Troubleshooting Fault Code t05-5121. (</qs3/pubsys2/xml/en/procedures/122/122-t05-5121.html>)

Last Modified: 16-Apr-2020
